



Campusmonk

Live Class – Diwali Special

Topic : Quadratic Equations - New TCS Type

Type 1:Concept Sheet

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Question 1 :

If α and β are the roots of $4x^2 + 3x + 7 = 0$, then the value of $\frac{1}{\alpha} + \frac{1}{\beta}$ is:

- a. $\frac{4}{7}$ b. $\frac{-3}{7}$ c. $\frac{3}{7}$ d. $\frac{-3}{4}$



Question 2:

If α and β are the roots of the equation $x^2 - x + 1 = 0$, then write the value $\alpha^2 + \beta^2$

- a. 1 b. -1 c. 0 d. None of these



Question 3:

If the equation $2x^2 - 7x + 12 = 0$ has root α and β , then the value of $\frac{\alpha}{\beta} + \frac{\beta}{\alpha}$

a. $\frac{97}{24}$

b. $\frac{7}{2}$

c. $\frac{1}{24}$

d. $\frac{7}{24}$



Type 2: Concept Sheet

Question 4:

One root of quadratic equation $x^2 - kx + 27 = 0$ is 3, then find the value of 'k'

a. 10

b. 12

c. -12

d. 16



Question 5:

**Find the value of k if one root of the equation:
 $x^2 - 9x + k = 0$ is twice the other root.**

- a. 18 b. 16 c. 12 d. 9**



Question 6:



**Find the value of k so that the sum of the roots of the equation
 $3x^2 + (2k + 1)x - k - 5 = 0$ is equal to the product of the roots.**

Type 3: Concept Sheet

Question 7(a):

If the sum of the roots of a quadratic equation is 1 and product of the roots is -20 . Find the quadratic equations.

- a. $x^2 - x - 20 = 0$
- b. $x^2 + x + 20 = 0$
- c. $x^2 + x - 20 = 0$
- d. $x^2 - x + 20 = 0$

Question 7(b):

Which of the following quadratic equation has roots -3 and -5

- a. $x^2 - 8x + 15 = 0$
- b. $x^2 - 8x - 15 = 0$
- c. $x^2 + 8x + 15 = 0$
- d. $x^2 + 8x - 15 = 0$

If α and β are the roots of equation :

$$x^2 - x + 1 = 0$$

then which equation will have roots α^3 and β^3

1. $x^2 - 2x + 1 = 0$
2. $x^2 + 2x + 1 = 0$
3. $4x^2 - 3x + 2 = 0$
4. $3x^2 - 5x + 2 = 0$



Q9. If α, β are roots of the equations $x^2 - 5x + 6 = 0$, then find the quadratic equation whose roots are $\frac{1}{\alpha}, \frac{1}{\beta}$

- a. $6x^2 - 5x + 1 = 0$
- b. $6x^2 + 5x + 1 = 0$
- c. $6x^2 - 5x - 1 = 0$
- d. $6x^2 + 5x - 1 = 0$

If α and β are the roots of equation :
 $x^2 - 3x + 2 = 0$

then quadratic equation whose roots are
 $(\alpha + 1)$ and $(\beta + 1)$ is ?

1. $x^2 - 5x + 6 = 0$
2. $x^2 + 5x + 6 = 0$
3. $x^2 - 5x + 2 = 0$
4. $x^2 - 5x - 6 = 0$

Type 5: Concept Sheet

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Type:

A quadratic equation is of the form $ax^2 + bx + c = 0$, where x is an unknown variable, a, b, c are constants and $a \neq 0$

$ax^2 + bx + c = 0$ is the standard form of quadratic equation.

Root of Quadratic Equation

It is the value of the unknown variable for which the quadratic equation holds true. A quadratic equation has two roots.

Nature of Roots

For equation $ax^2 + bx + c = 0$, expression $b^2 - 4ac$ is called Discriminant and denoted by D .

Value of Discriminant	Nature of roots
$D < 0$	Unequal and imaginary
$D = 0$	Real and equal
$D > 0$	Real and unequal

$D > 0$ and is a perfect square	Real, unequal and rational
$D > 0$ and not a perfect square	Real, unequal and irrational

Solution of Quadratic Equation	
$x^2 - 6x + 5 = 0$	
Factorization - Not conveniently applicable for all quadratic equations - Simplest method Example: $x^2 - 6x + 5 = 0$; Standard form $x^2 - 6x + 5 + 4 - 4 = 0$ $x^2 - 6x + 9 = 4$; Rearrange terms	Completing square Add and subtract square of half coefficient of x Example: $x^2 - 6x + 5 = 0$; Standard form $x^2 - 6x + 5 + 4 - 4 = 0$ $x^2 - 6x + 9 = 4$; Rearrange terms
Quadratic formula - Formula used to calculate, $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$ - Applicable for all equations Example: $x^2 - 6x + 5 = 0$ Compare with $ax^2 + bx + c = 0$ To get $a = 1, b = -6, c = 5$ $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$	

Question 1: Which of the following quadratic Equations has real roots?

1. $5x^2 - 2x + 2 = 0$
2. $3x^2 - 4x + 2 = 0$
3. $4x^2 - 3x + 2 = 0$
4. $3x^2 - 5x + 2 = 0$

Question 2 :

For what positive values of m , does the equation $(3m+1)x^2 + 2(m+1)x + m = 0$ have real and equal roots ?

TITA